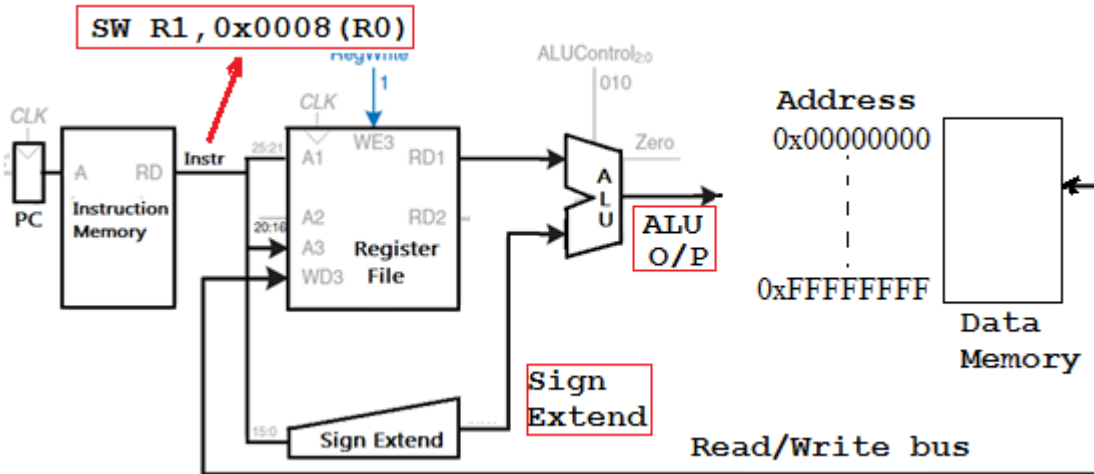


## CA-Home Assignment 06

- Answer the following questions based on the single-cycle data path of the MIPS system shown in **Figure 6**. Assume that the registers **R0** and **R1** have initialized with **0x00000007** and **0xC123FAE7**.



- Show the O/P of the Sign Extend unit (in hexadecimal).
  - Show the O/P of the ALU (in hexadecimal).
  - Show the **byte values** in the data memory (DM) with their corresponding 32-bit physical addresses after the **SW** execution.
- Based on the following MIPS instruction sequence, determine the **32-bit values** (in hexadecimal) of the registers **R1** and **R2** after the instruction execution (assume that the register **R0** is initialized with **0x00000001**).

- 1). `ADDI R1, R0, 0x0004`
- 2). `ADDI R2, R1, 0x0005`
- 3). `SLL R1, R1, 0x0001`
- 4). `BEQ R1, R2, TARGET`
- 5). `ADD R2, R2, R1`
- 6). `TARGET: SUB R2, R2, R1`

- Convert the following high-level code section to MIPS instructions (where **q** and **h** are integer variables):

```

if (g > h)
    g = g + h;
else
    g = g - h;

```

4. Consider the MIPS system executing the following instruction as part of a benchmark test: **LW R1, 0x0006(R0)**. Assume the register **R0** is initialized with **0x00000006**. A section of the MIPS's byte-addressable **data memory (DM)** is shown in **Figure 6**. Find out the value of **R1** after the execution of the **LW** instruction.

| Memory Address | Memory Word |
|----------------|-------------|
| 0x00000008     | 0x23        |
| 0x00000009     | 0x5A        |
| 0x0000000A     | 0x1E        |
| 0x0000000B     | 0xA0        |
| 0x0000000C     | 0x5B        |
| 0x0000000D     | 0x4C        |
| 0x0000000E     | 0x67        |
| 0x0000000F     | 0x32        |
| 0x00000010     | 0xAF        |

5. Consider the following **high-level language code** running on a MIPS system. **Show its MIPS assembly code translation, including complete memory operations with the data memory (DM)** for the variable allocation. The size of the integer value in each variable is **32-bit**. The **32-bit memory address** of the first byte of the variable, **a**, is **0xA000F004**. The variables **b** and **c** use the consecutive byte addresses for their values in **DM**. Use the 32-bit registers **R1**, **R2**, and **R3** for **a**, **b**, and **c** in memory and arithmetic operations, and register **R0** as the base register, initialized to **0xA0000001**.

```
int a = 25;
int b = 15 - a;
int c = b + 30;
```

6.